Editing

Good writing takes time and effort.
It should not be rushed.

Fish 290

Thursday March 2rd

Peer editing

Your edits are due Friday at 11:59 pm

3-step editing process:

1. **Fine details:** Read through the paper. Use Track Changes to mark any unclear statements, give comments, make edits
2. **Overall structure:** Go through checklist (we will make this today). Check that each section/paragraph is doing its job.
3. **Big picture:** See Table 1 of "10 Simple Rules for Structuring Papers." Are there signs that the 10 simple rules are violated? If so, what do they need to change?

1. Fine details

<u>The reader does not know what you know.</u>

Signs you wrongly assumed your reader is also in your brain:

1. Justification for why your study is needed is missing or weak in the Introduction
2. Important context is missing (study site, basics of organism's biology relevant to study)
3. Pronoun-heavy writing ("It was taken to the lab…" "Temperature affected them…")

This is why it is necessary to get feedback from someone else!

1. Fine details

Brevity: always try to communicate your message in as few words as possible. ***Every sentence must serve a purpose.***

Why be brief?

- Shorter writing is clearer writing
- **Fluff** competes for your reader's attention and dilutes the main message
- Fluff is transparent and gives poor impression of your work
- Many journals and grants have strict length limits

Read "Brevity" in Canvas if you need help cutting!

Don’t take a paragraph to say what can be said in one sentence.

There are several factors that contribute to coral bleaching. Two of those factors are increased temperature and decreasing pH. When temperatures get warmer, this can cause corals to become bleached. Similarly, when the water gets more acidic, this stress can also lead to coral bleaching. Exposure to these factors can be both long term, intermediate, and short term. Thus, there are a multitude of factors to consider when trying to understand the root cause of coral bleaching, and researchers should place high emphasis on the combined effects of increased temperature and ocean acidification.

Exposure to warmer temperatures and acidic waters can lead to coral bleaching (Martinez and Colins, 2021).

1. Fine details

Common redundancies to avoid (eliminate words in parentheses):

(already) existing
(alternative) choices
At (the) present (time)
(basic) fundamentals
(completely) eliminate
(continue to) remain
(currently) being
(currently) underway
(empty) space
Had done (previously)

introduced (a new)
mix (together)
never (before)
none (at all)
now (at this time)
period (of time)
(still) persists
start (out)

Apex predators are important for the ocean.

Without substance.

A considerable amount of research has shown that apex predators are vitally important for the ocean for a variety of reasons.

Without substance, but now with fluff.

Reductions in the abundance of apex predators due to overfishing can cause **trophic cascades** that lead to declines in the **quality of marine habitats** (Munoz et al 2011) and **decreases in species richness** (Yang et al. 2023).

Precise. Nice.

A considerable amount of research has shown that apex predators are vitally important for the ocean for a variety of reasons. First, removing predators can cause trophic cascades that lead to the decline in the quality of habitats (Munoz et al. 2011). In addition, when top predators are reduced, studies have also reported decreases in species richness in the ecosystem overall (Yang et al. 2023).

Reductions in the abundance of apex predators due to overfishing can cause trophic cascades that lead to declines in the quality of marine habitats (Munoz et al 2011) and decreases in species richness (Yang et al. 2023).

Avoid jargon and seek simplicity, brevity, and clarity

Jargon or complex version	Simple, brief, and clear version
A considerable amount of	much
A considerable number of	many
A decreased amount of	less
A decreased number of	fewer
A majority of	most
A number of	many
A small number of	few
Absolutely essential	essential
Accounted for by the fact that	because
Adjacent to	near
Along the lines of	like
An adequate amount of	enough
An example of this is the fact that	for example

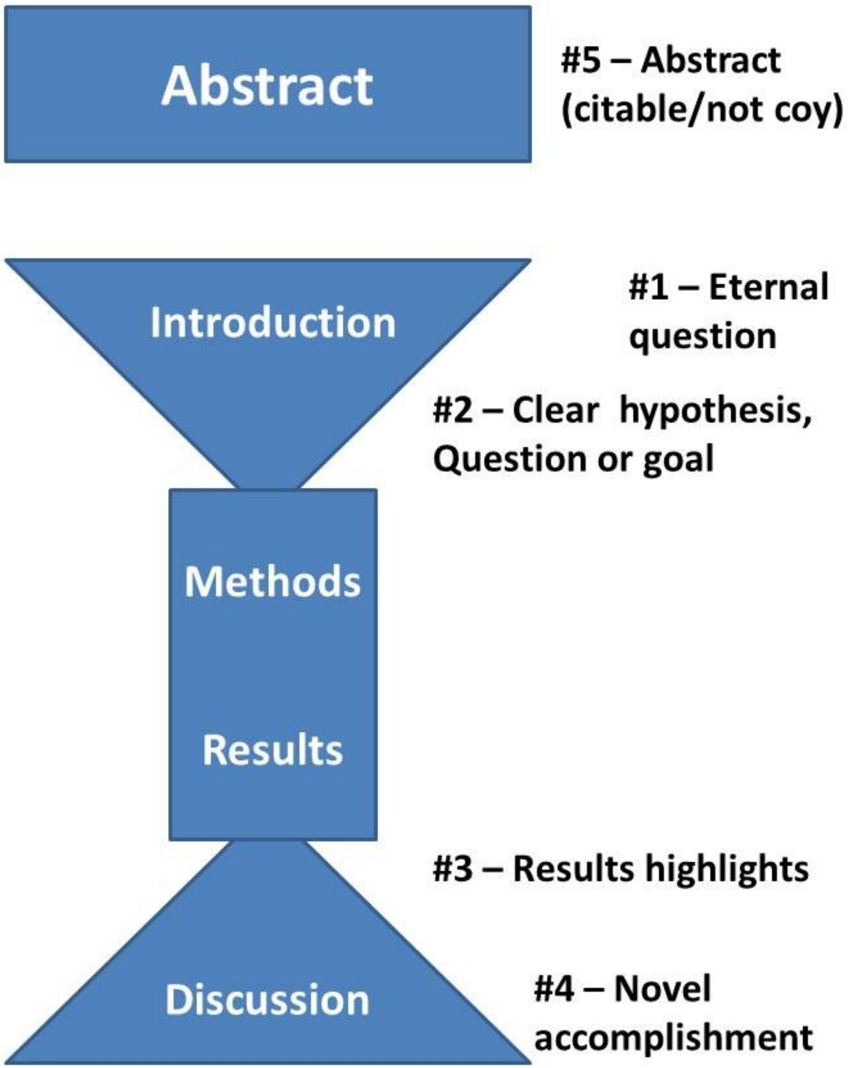
1. Fine details

Brevity: always try to communicate your message in as few words as possible. ***Every sentence must serve a purpose.***

I challenge to you cut 10-20% of the words in your first draft.

Read “Brevity” in Canvas if you need help cutting!

2. Overall structure



Every sentence must serve a purpose.

Every paragraph must also serve a purpose.

2. Overall structure

Every paragraph must serve a purpose.

Use signposts. A signpost is a familiar phrase that marks key statements in your paper. Signposts are needed to guide the reader though technical writing.

Examples:

- Intro: "In this study, I tested the relationship between…"
- Results/Discussion: "Overall, our experiments showed that…"
- Discussion: "In conclusion, we found that…"

3. Big picture (or, what makes a paper “good”?)

PLOS | COMPUTATIONAL BIOLOGY

EDITORIAL

Ten simple rules for structuring papers

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3. Big picture (or, what makes a paper “good”?)

Table 1. A summary of the ten rules and how to tell if they are being violated.

Rule	Sign it is violated
1: Focus on one big idea	Readers cannot give 1-sentence summary.
2: Write for naive humans	Readers do not “get” the paper.
3: Use context, content, conclusion structure	Readers ask why something matters or what it means.
4: Optimize logical flow	Readers stumble on a small section of the text.
5: Abstract: Compact summary of paper	Readers cannot give the “elevator pitch” of your work after reading it.
6: Introduction: Why the paper matters	Readers show little interest in the paper.
7: Results: Why the conclusion is justified	Readers do not agree with your conclusion.
8: Discussion: Preempt criticism, give future impact	Readers are left with unanswered criticisms and/or questions on their mind.
9: Allocate time wisely	Readers struggle to understand your central contribution despite your having worked hard.
10: Iterate the story	The paper’s contribution is rejected by test readers, editors, or reviewers.

A helpful checklist compiled from previous lectures:

1) Does the Introduction start broadly and end with the objectives?
2) Does the Introduction give only enough of the approach to understand what is to come – some information but not too much detail?
3) Does the Methods section reveal what was done, where the data came from, and any other special features, but not disclose the results? Sample size, period of record, and such things are often in the Methods if they are needed to understand the study design, though they may appear in the Results.
4) Does the Results section present all the available data needed to test all the hypotheses but no extraneous information?
5) Does the Results section avoid interpretation, apologies and other junk?
6) Are the sample sizes, treatments and controls, replicates clear?
7) Do all figures and tables have complete, self-explanatory captions?
8) Are figures and tables easy to understand?
9) Does the Discussion connect the key results with the hypotheses?
10) Does the Discussion integrate the findings with work by others?
11) Does the Discussion end with an appropriately broad connection back to the themes of the Introduction?

Today:

1. Finish your peer reviews. Submit the "Guide to Peer-Review" on canvas, separate from the actual word document you are reviewing. Both are due tomorrow at midnight.

2. Work on your presentations. We will have *some* time on Tuesday to do this as well.